Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = \underset{\nwarrow}{V} + \underset{\nearrow}{\sqrt{2V}}\,\eta \;\longleftarrow\; \begin{array}{c}\text{Gaussian noise}\\ \langle \eta(t)\eta(t')\rangle \sim \delta(t-t')\end{array}$$

Drift = Diffusion

SPRT mean decision time (Siggia & Vergassola 2012)

$$\langle T_{\mathrm{dec}}\rangle = \frac{K}{V}\tanh\left(\frac{K}{2}\right)$$

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty}\frac{1}{t}\left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2}\right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P($$

$$|c)$$